MODULE

8

Storage & Transportation

Compressed · liquid · underground · hydrides · pipelines · trailers · ships

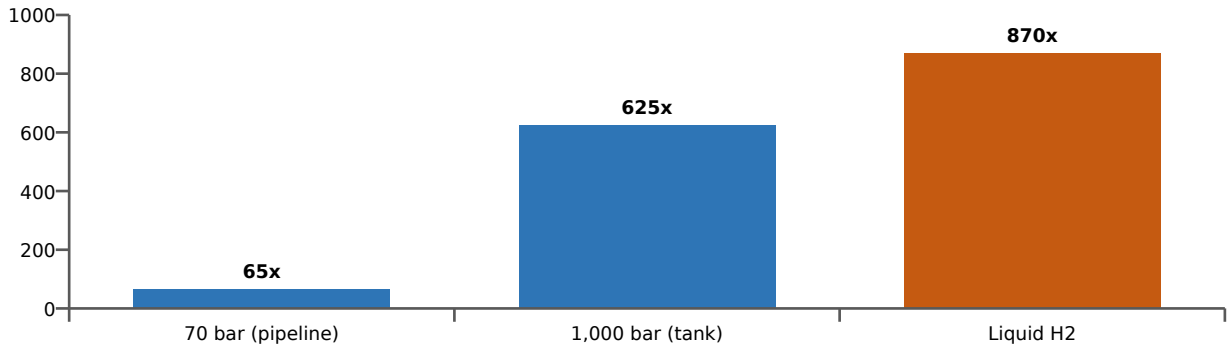
8.1 Storage methods

Method	How it works	Typical use	Advantages	Limitations
Compressed gas	Steel or composite vessels; vehicle tanks at 350 or 700 bar	Vehicles, refuelling stations, plant buffers	Simple, fast fill/discharge	Low volumetric density; compression energy
Liquid H_2	Cryogenic, vacuum-insulated tanks at –253 °C	Bulk delivery, space, shipping	High density per tank	Liquefaction energy; boil-off
Underground (salt cavern)	Gas stored in solution-mined caverns	Seasonal / large-scale storage	Very large volume at low cost per kg	Geology-dependent; few sites
Metal hydrides	H_2 absorbed into metal alloy lattice	Small stationary, submarines, niche	Low pressure, compact, safe	Heavy; heat management
Chemical carriers (NH_3, LOHC, methanol)	H_2 bound in a liquid; released by cracking/dehydrogenation	Export / long distance	Liquid handling with existing assets	Conversion & reconversion losses

8.2 Pressure vessel types

Type	Construction	Notes
Type I	All-metal (steel)	Heaviest, cheapest; stationary
Type II	Metal liner, hoop-wrapped composite	Stationary
Type III	Metal (aluminium) liner, fully wrapped composite	Lighter; transport
Type IV	Polymer liner, fully wrapped carbon fibre	Lightest; 700 bar car tanks

Volume reduction vs. atmospheric-pressure H2 (x times smaller)



Source: IRENA (2020) citing BNEF (2019) [R1].

8.3 Transportation

Mode	Description	Best for
Pipelines	Dedicated H_2 lines (design code ASME B31.12) or repurposed gas lines after integrity assessment	Large, steady flows over short–medium distance
Tube trailers	Bundles of high-pressure cylinders on a truck	Small–medium customers, refuelling stations
Liquid H_2 tankers	Cryogenic road tankers	Larger volumes over longer road distances

Mode	Description	Best for
Ships	LH_2 carriers (first demonstration ship operated Australia → Japan) or ammonia/LOHC tankers	Intercontinental trade
On-site production	Electrolyser at the point of use	Avoids transport altogether

Engineering challenge – embrittlement in transport
High-strength steels can lose ductility in hydrogen (Module 10). Pipeline repurposing therefore needs material testing, lower design stress and fatigue checks under pressure cycling.

Quiz & revision – Module 8

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Vehicle tank pressures commonly used:
a) 35 & 70 bar **b)** 350 & 700 bar **c)** 1,000 & 2,000 bar **d)** 10 & 20 bar

Q2. Which tank type uses a polymer liner with full carbon-fibre wrap?
a) I **b)** II **c)** III **d)** IV

Q3. Best option for seasonal, very large storage:
a) Tube trailers **b)** Salt caverns **c)** Metal hydride **d)** Type IV tanks

Q4. Design code for hydrogen piping & pipelines:
a) ASME B31.12 **b)** API 650 **c)** IS 456 **d)** ISO 9001

Q5. Main drawback of chemical carriers like ammonia:
a) Toxic & conversion losses **b)** Too light **c)** Need 700 bar **d)** Cannot be shipped

Answer key
Q1 – b: ; **Q2 – d**: ; **Q3 – b**: ; **Q4 – a**: ; **Q5 – a**: cracking back to H2 costs energy